Freeing ω_m across multipoles

A CMB diagnostic of ΛCDM: model, binning, mock-data MCMC and code benchmarks

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All results use mock (noiseless) Planck-like data. Chains: 8×10^5 steps per run on Hive.

The idea: let ω_m depend on the multipole

- Split $\ell = 2 \ldots 2508$ into N bins; bin $0 = [2, \ell_{s,1}]$, bin $i = (\ell_{s,i}, \ell_{s,i+1}]$.
- In bin i every spectrum (TT, TE, EE, $\phi\phi$) is the ΛCDM one with

$$\omega_{\rm cdm} = \omega_{m,i} - \omega_b,$$

  all other parameters ($\omega_b, h, \tau, n_s, A_s$) shared.
- $\Lambda\text{CDM} \Leftrightarrow \omega_{m,0} = \cdots = \omega_{m,N-1}$.
  Test: $\chi^2 = \bar{\Delta}^T C_\Delta^{-1} \bar{\Delta}$ on $\Delta_i = \omega_{m,i} - \omega_{m,0}$, $N-1$ d.o.f.
- The bin edges $\ell_{s,i}$ are fixed in each run (not sampled).

What it is: a scale-dependent consistency test. Physics that changes the damping tail differently from the acoustic peaks shows up as $\omega_{m,i} \neq \omega_{m,j}$.

What it is not: a cosmology. There is no single background, so Ω_m, $H(z)$, σ_8, r_s are undefined $\Rightarrow$ CMB-only, no BAO/SN.

By construction: spectra jump at the bin edges (slide 13).

Three questions for this talk: (1) where to put the edges, (2) does the binned fit detect a step it was not designed for, and (3) can we trust the fast spectra (CosmoPower) we use for the chains?

Pipeline

Sampler: MontePython 3.6.1 (Metropolis–Hastings), shared by both pipelines.

- 8 chains $\times$ 10^5 steps per run, 30% burn-in.
- Proposal: inverse Fisher matrix of the same likelihood.
- Convergence: $R - 1 \leq 0.002$ for every parameter of every run; acceptance 0.14–0.30.

Parameters: $100\,\omega_b$, h, $\tau_{\rm reio}$, n_s, $\ln(10^{10} A_s)$, $\omega_{m,0..N-1}$; flat priors ($h \in [0.6, 0.75]$, $\omega_{m,i} \in [0.05, 0.25]$).

Two spectra back-ends, same interface (a `classy` look-alike, so MontePython is unchanged):

`with_cosmopower`	CosmoPower CP_paper emulators (TT, EE, TE, $\phi\phi$; lensed; $\ell \leq 2508$), one batched call per bin. $\sim$0.1 s/step.
`with_class`	real CLASS v3.4, one run per bin (defaults: massless ν, linear lensing). $\sim$3.5 s/bin/step.

Each pipeline makes its own mock with `make_fiducial.py` (same format as MontePython's writer).

Likelihood: MontePython `fake_planck_realistic`

Exact full-sky Gaussian (Wishart) likelihood per multipole, scaled by $f_{\rm sky}$:

$$-2\ln\mathcal{L} = \sum_{\ell=2}^{2508}(2\ell+1)f_{\rm sky}\left[\mathrm{Tr}\left(\mathsf{C}_\ell^{-1}\hat{\mathsf{C}}_\ell\right)+\ln\frac{\det\mathsf{C}_\ell}{\det\hat{\mathsf{C}}_\ell}-n\right]$$

C_ℓ = theory + noise, $\hat{\mathsf{C}}_\ell$ = mock + noise.

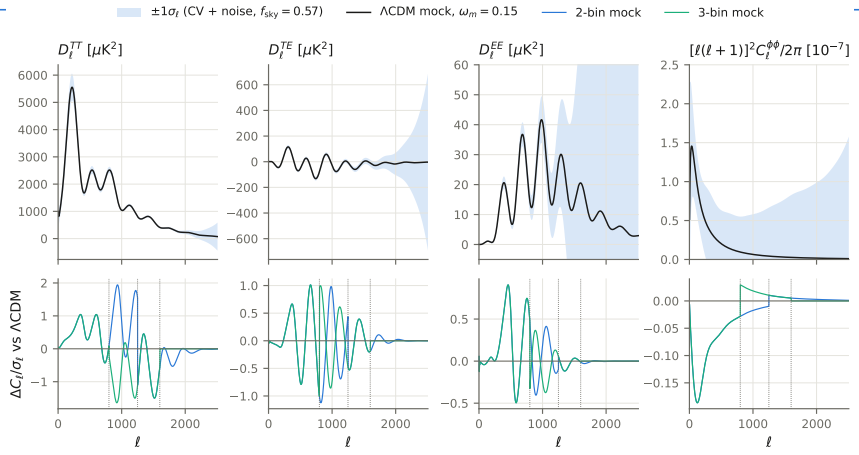
- Fields: T, E and the lensing deflection d (lensing extraction); T–d correlation neglected.
- Noise: Planck 100 + 143 GHz (Brinckmann, Hooper et al. 2018); $f_{\rm sky} = 0.57$.
- $\ell_{\rm max} = 2508$ (emulator range; MontePython default is 3000).
- Mocks are *noiseless*: $\chi^2_{\rm min} = 0$ when the model contains the truth.

mock	$\omega_m(\ell)$	steps at
ΛCDM	0.15	—
2-bin	0.14 | 0.16	1250
3-bin	0.14 | 0.16 | 0.15	800, 1600

Common: $\omega_b = 0.0224$, $h = 0.674$, $\tau = 0.054$, $n_s = 0.965$, $\ln(10^{10}A_s) = 3.044$.

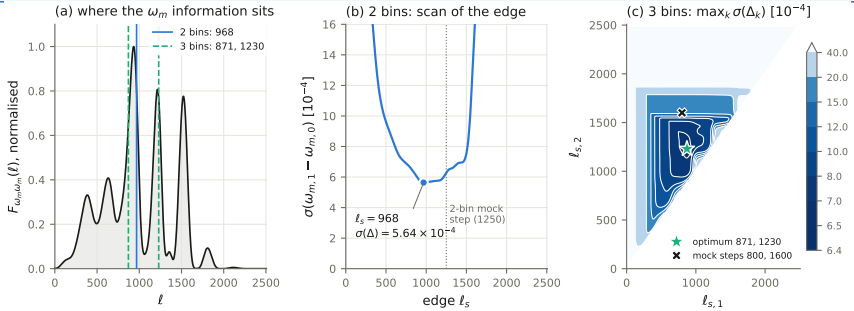
The injected steps (1250; 800, 1600) are deliberately *not* at the edges we fit with (968; 871, 1230).

The mocks, and how far the binned ones are from ΛCDM



- Top: ΛCDM mock with the per-ℓ 1σ of our likelihood (CV + noise); TT is noise dominated above $\ell \approx 1500$.
- Bottom: binned mocks minus ΛCDM, in units of σ_ℓ: up to 2σ *per multipole*, over hundreds of multipoles.

Choice of binning: Fisher forecast

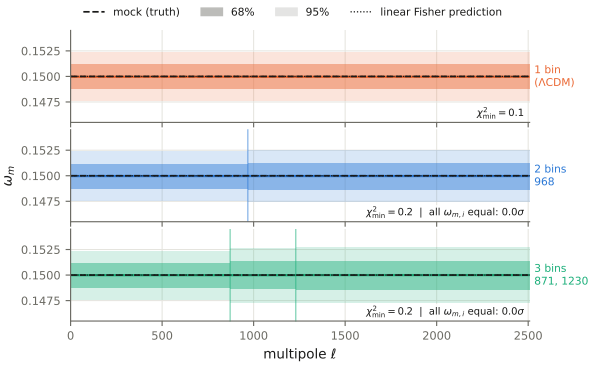


- Objective: minimise $\max_k \sigma(\omega_{m,k+1} - \omega_{m,k})$, marginalised over ω_b, h, τ, n_s, A_s and the other bins.
- Fisher of this likelihood (TT, TE, EE + d), derivatives with real CLASS (5-point stencil; $< 0.6\%$ change when steps are halved).
- Since $\partial C_\ell/\partial\omega_{m,i} = \partial C_\ell/\partial\omega_m \mathbb{1}[\ell \in i]$, any binning is a cumulative sum $\Rightarrow$ exhaustive scans are cheap.

N	edges	$\sigma(\omega_{m,i})$ [10^{-3}]	$\sigma(\Delta)$ [10^{-3}]
1	—	1.21	—
2	968	1.22, 1.29	0.56
3	871, 1230	1.22, 1.36, 1.39	0.65, 0.65

Optimum found by exhaustive scan ($N = 2$), differential evolution with 3 seeds + brute force on a 5-ℓ grid ($N = 3$): all agree. Rounding to multiples of 25 costs $< 0.1\%$.

MCMC 1: ΛCDM mock — the null test



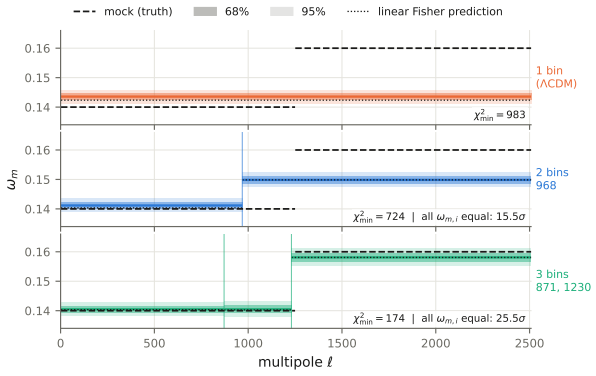
Fisher vs MCMC errors (ΛCDM mock):

	Fisher	MCMC
1 bin: $\sigma(\omega_m)$	1.21	1.21
2 bins: $\sigma(\omega_{m,i})$	1.22, 1.29	1.22, 1.29
$\sigma(\Delta)$	0.56	0.56
3 bins: $\sigma(\omega_{m,i})$	1.22, 1.36, 1.39	1.22, 1.37, 1.39
$\sigma(\Delta_k)$	0.65, 0.65	0.64, 0.64

(units of 10^{-3})

- All bins recover $\omega_m = 0.150$; Δ consistent with 0 (test: 0.0σ).
- $\chi^2_{\rm min} = 0.1$–0.2 (noiseless mock).
- The other parameters do not care how many bins we free: next slide, top row.

MCMC 2: 2-bin mock, fitted with misplaced edges

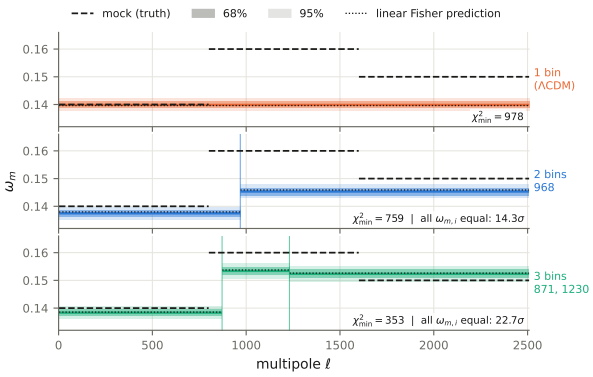


Truth: step 0.14 → 0.16 at $\ell = 1250$. Fitted edges: 968 (2 bins), 871 and 1230 (3 bins).

- 2 bins: $\omega_{m,1} - \omega_{m,0} = 0.0087 \pm 0.00056$ ⇒ ΛCDM rejected at 15.5σ, although the edge is 280 multipoles off.
- 3 bins: the jump lands between bins 1 and 2 ($\Delta_{21} = 0.0176 \pm 0.0007$); bins 0 and 1 agree. 25.5σ.
- Bins that straddle the step return an information-weighted average: 0.1499 for $\ell \in (968, 2508]$.
- $\chi^2_{\rm min}$: 983 → 724 → 174.

The binned model picks up the signal even when its edges do not match the truth.

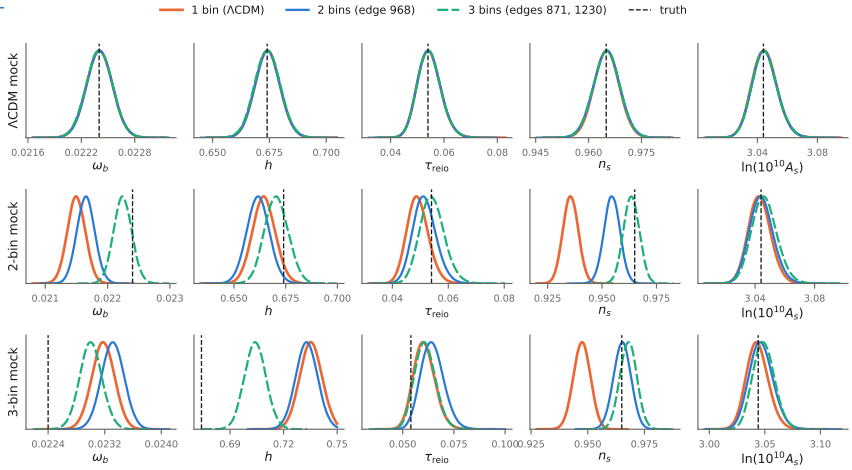
MCMC 3: 3-bin mock (0.14 | 0.16 | 0.15 at 800, 1600)



- 2 bins: $\Delta = 0.0079 \pm 0.00055$, 14.3σ.
- 3 bins: $\Delta_{10} = 0.0151 \pm 0.0007$, $\Delta_{21} = -0.0011 \pm 0.0007$; ΛCDM rejected at 22.7σ.
- The rise at 800 is caught by the 871 edge; the small drop at 1600 (0.16→0.15) falls inside the last bin and is averaged to 0.152.
- $\chi^2_{\rm min}$: 978 → 759 → 353; with these edges the model cannot reach the truth, so $\chi^2_{\rm min}$ stays large.

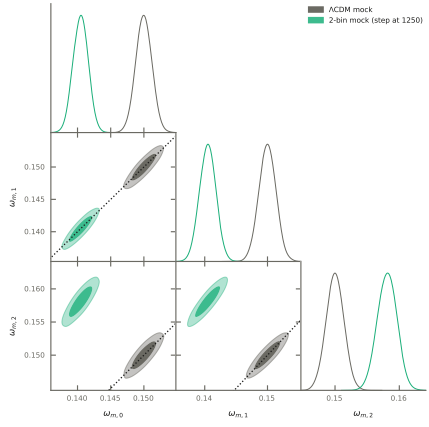
Dotted lines: linear Fisher prediction of the best fit (slide 12).

ΛCDM parameters: insensitive to extra bins, biased by a missed step

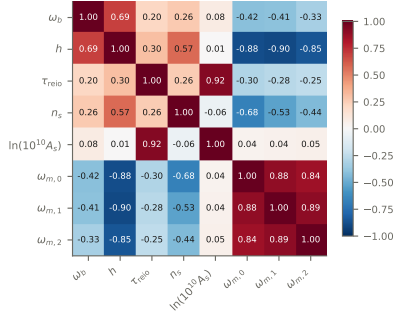


- ΛCDM mock (top): 1, 2 and 3 bins give the same posteriors (shifts $\leq 0.12\sigma$; $\sigma(h)$ 0.0055 → 0.0057). Freeing ω_m per bin costs nothing.
- Binned mocks: the 1-bin (ΛCDM) fit absorbs the step into n_s (-8.3σ) and ω_b (-6.3σ), or into h ($+10\sigma$, close to the prior edge 0.75). More bins remove most of the bias (2-bin mock, 3 bins: all $|\mathrm{shift}| \leq 1.1\sigma$).

Bin posteriors and correlations



3-bin fit. Dotted: $\omega_{\boldsymbol{m},i} = \omega_{\boldsymbol{m},j}$ (ΛCDM).



Correlation matrix, 3-bin fit to the ΛCDM mock.

- Bins are strongly correlated with each other ($\rho = 0.84$–0.89) and with h ($\rho \approx -0.88$).
- The common mode goes into h; the differences are what the data pin down: $\sigma(\Delta) = 0.56 \times 10^{-3}$ vs $\sigma(\omega_{\boldsymbol{m},i}) = 1.2$–$1.4 \times 10^{-3}$.

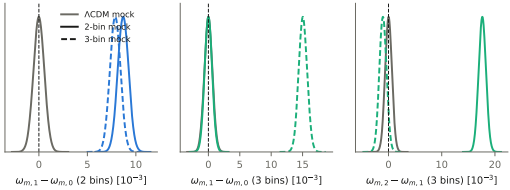
Comparison with the model: a linear Fisher prediction reproduces the chains

Best fit of the binned model to a mock with $\omega_m(\ell)$, to first order:

$$\delta\hat{p} = F_{\rm fit}^{-1}\, b, \qquad b_a = \sum_\ell F_\ell^{a\,\omega_m}\, \delta\omega_m(\ell),$$

$$\chi^2_{\rm min} = \sum_\ell F_\ell^{\omega_m\omega_m}\, \delta\omega_m(\ell)^2 - b^T F_{\rm fit}^{-1} b.$$

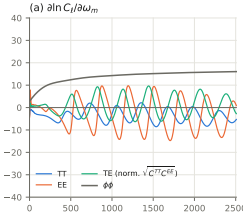
mock	fit	$\chi^2_{\rm min}$ linear	MCMC
2-bin	1 bin	1010	983
	2 bins	726	724
	3 bins	164	174
3-bin	1 bin	945	979
	2 bins	748	759
	3 bins	354	353

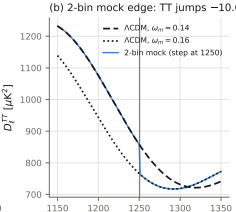


Δ posteriors. Colour = fitted binning (blue: 2 bins, green: 3 bins).

- Bin values agree with MCMC to $\lesssim 0.001$ (dotted lines on slides 8–9), ΛCDM parameters to $\lesssim 0.5\sigma$.
- The chains are understood: each fitted bin returns the Fisher-weighted mean of $\omega_m(\ell)$ over its range, and the rest is absorbed by n_s, ω_b, h.

Derivatives and the discontinuity at the edges





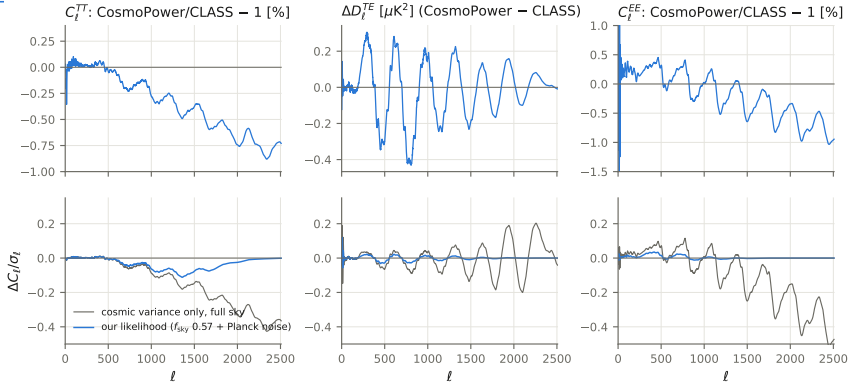
Jump in C_ℓ across an edge (real CLASS) and the error it puts in a central-difference $\partial C_\ell / \partial \ell$ at the edge:

	TT	EE	TE†	$\phi\phi$
$1\sigma(\Delta)$ step (5.6×10^{-4}) at 1250				
jump	**−0.31%**	**+0.24%**	**−0.29%**	**+0.8%**
natural $\ell \to \ell+1$	**−0.54%**	**+0.33%**	**−0.35%**	**−0.5%**
err. on $\partial C/\partial \ell$	**+28%**	**+35%**	**+42%**	**−88%**
mock step 0.14 → 0.16 at 1250				
jump	**−10.6%**	**+9.0%**	**−10.2%**	**+35%**
err. on $\partial C/\partial \ell$	**×9**	**×9**	**×19**	**×36**

† **TE relative to $\sqrt{C^{TT} C^{EE}}$. Same size at 871, 968, 1230.**

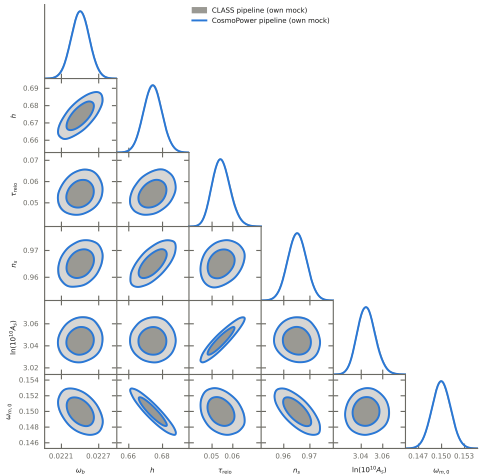
- $|\partial \ln C_\ell / \partial \omega_m| \sim 5$–$15$, so a step $\delta\omega_m$ makes a jump $\approx \partial \ln C / \partial \omega_m \cdot \delta\omega_m$; at a just-detectable step it is about half the normal change between neighbouring multipoles.
- Our likelihood and the Fisher use C_ℓ at fixed ℓ (and $\partial/\partial\theta$ only) ⇒ unaffected. Plik, CamSpec, Commander and Planck lensing never take $\partial C_\ell / \partial \ell$; SPT-3G (aberration, super-sample lensing) and Cobaya's CMBlikes aberration do ⇒ put edges on bandpower edges there.

Benchmark 1: spectra, CLASS vs CosmoPower



- At the mock point ($\omega_{\boldsymbol{m}} = 0.15$): CosmoPower is low in the damping tail, TT -0.3% at $\ell = 1000$, -0.7% at 2000–2500; EE up to -1%.
- Per multipole in our likelihood: $\leq 0.11\sigma_\ell$ (TT), $\leq 0.04\sigma_\ell$ (TE, EE). Cosmic-variance-limited: up to 0.4–0.5σ.
- Not CLASS settings: `cl_ref.pre` changes CLASS by $\leq 0.08\%$; CAMB (massless ν, linear lensing) shows the same gap. The emulators were trained on CAMB over a very wide box; the size matches the CosmoPower paper's full-box accuracy ($\sim 0.2\sigma_{\rm CMB}$, App. C).
- ℓ indexing checked: best-fit shift $|d| \leq 0.27$; an injected 1-multipole shift is recovered.

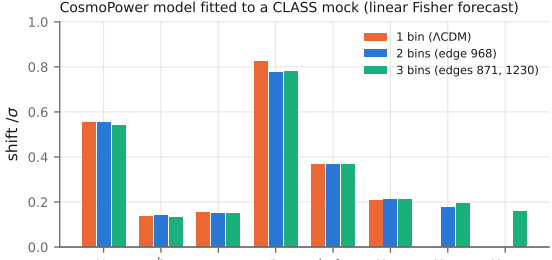
Benchmark 2: MCMC level



ΛCDM mock, 1 bin. Each pipeline fits its own mock.

Same pipeline, different spectra code (each with its own mock): posteriors identical; mean shifts $\leq 0.02\sigma$, error ratios 0.98–1.00.

Cross test (forecast): CosmoPower model fitted to a CLASS mock — what real data would see.



- Emulator error is absorbed by n_s ($+0.8\sigma$), ω_b ($+0.55\sigma$), A_s ($+0.37\sigma$); χ^2 6.4 → 0.5 after refit.
- Spurious $\Delta\omega_m$ between bins: $\leq 0.06\sigma$. (Supersedes our earlier single-parameter estimate of $+0.001$.)

Summary and next steps

Where we are

- Binned-ω_m model in MontePython with two interchangeable back-ends; mock Planck-like likelihood (TT, TE, EE + lensing).
- Fisher-optimal edges: 968 (2 bins); 871, 1230 (3 bins). $\sigma(\Delta) = 0.56$–0.65×10^{-3}, confirmed by MCMC.
- Freeing ω_m per bin leaves the ΛCDM posteriors unchanged on a ΛCDM mock.
- Misplaced bins still detect injected steps at 14–26σ; a ΛCDM fit instead hides them in n_s, ω_b, h.
- The chains match a linear Fisher prediction.
- CosmoPower vs CLASS: identical posteriors on their own mocks; forecast emulator bias on real data is in the ΛCDM parameters ($\leq 0.8\sigma$), not in $\Delta\omega_m$.

Next

- Cross-pipeline MCMC: CosmoPower chains on the CLASS mock (check the forecast bias).
- CLASS counterparts of the binned runs (same folders).
- Sensitivity: smallest detectable step vs its position; more bins (4–10).
- Real data: Planck likelihoods via clik with CLASS (or a narrow-box emulator retrained on CLASS); edges on bandpower edges.

Backup

Backup: convergence and best fits (CosmoPower pipeline)

mock	fit	acceptance	max $R-1$	$\chi^2_{\rm min}$	ΛCDM test
ΛCDM	1 bin	0.25	0.0002	0.09	—
	2 bins	0.25	0.0007	0.17	0.0σ
	3 bins	0.14	0.0015	0.17	0.0σ
2-bin	1 bin	0.25	0.0004	983.0	—
	2 bins	0.24	0.0007	724.1	15.5σ
	3 bins	0.14	0.0013	173.5	25.5σ
3-bin	1 bin	0.30	0.0006	978.5	—
	2 bins	0.26	0.0008	759.5	14.3σ
	3 bins	0.15	0.0016	353.0	22.7σ
CLASS pipeline, ΛCDM mock, 1 bin		0.26	—	0.08	—

8 chains $\times 10^5$ steps each, 30% burn-in; $R-1$: multiplicity-weighted Gelman–Rubin per parameter, worst parameter shown. ΛCDM test: Gaussian χ^2 of $(\Delta_1, \ldots, \Delta_{N-1})$ against 0, converted to a two-sided significance.

Backup: marginal constraints, all runs

mock	fit	$10^2\omega_b$	h	τ	n_s	$\ln 10^{10} A_s$	$\omega_{m,i}$
truth		2.240	0.674	0.054	0.965	3.044	
ΛCDM	1	2.240(15)	0.6742(55)	0.0545(44)	0.9651(36)	3.045(8)	0.1500(12)
	2	2.241(15)	0.6742(55)	0.0545(43)	0.9651(37)	3.045(8)	0.1500, 0.1500
	3	2.240(15)	0.6741(57)	0.0545(44)	0.9651(37)	3.045(8)	0.1500, 0.1500, 0.1500
2-bin	1	2.150(14)	0.6645(55)	0.0491(38)	0.9356(35)	3.043(8)	0.1436(12)
	2	2.165(14)	0.6616(55)	0.0515(41)	0.9545(37)	3.044(8)	0.1412, 0.1499
	3	2.224(15)	0.6705(57)	0.0543(44)	0.9635(36)	3.046(9)	0.1405, 0.1405, 0.1581
3-bin	1	2.318(16)	0.7352(61)	0.0606(52)	0.9474(37)	3.043(10)	0.1399(12)
	2	2.331(16)	0.7329(61)	0.0645(56)	0.9652(38)	3.047(11)	0.1375, 0.1453
	3	2.300(16)	0.7041(60)	0.0610(51)	0.9679(37)	3.049(10)	0.1384, 0.1534, 0.1524

Mean (error on the last digits). $\sigma(\omega_{m,i})$: 1.2–1.5×10^{-3}, see slide on errors. Binnings: 2 bins = edge 968; 3 bins = edges 871, 1230. 3-bin mock, 1–2 bins: h is within 3σ of the prior edge 0.75.

Backup: what the emulator error is, and is not

Emulator / reference -1 [%], TT unless noted, at $\omega_b = 0.0224$, $\omega_{\rm cdm} = 0.12$, $h = 0.674$, $\tau = 0.054$, $n_s = 0.965$, $\ln 10^{10} A_s = 3.044$:

reference	$\ell = 30$	500	1000	1500	2000	2500
CLASS 3.4, massless ν, linear lensing	$+0.03$	-0.02	-0.22	-0.34	-0.71	-0.78
EE	$+0.46$	$+0.23$	-0.02	-0.61	-0.28	-0.95
CLASS 3.4, $m_\nu = 0.06$ eV	$+0.01$	-0.26	$+0.15$	$+0.53$	-0.70	$+0.41$
CLASS 3.4, halofit lensing	$+0.03$	-0.01	-0.22	-0.27	-0.91	-1.77
CLASS 2.9.4, massless ν, linear	$+0.03$	$+0.00$	-0.20	-0.31	-0.55	-0.52
CAMB 2.0.4, $m_\nu = 0$, linear lensing	$+0.05$	-0.10	-0.19	-0.31	-0.64	-0.56
CAMB 2.0.4, $m_\nu = 0$, nonlinear lensing	$+0.04$	-0.08	-0.19	-0.23	-0.82	-1.49
CLASS default vs `cl_ref.pre` (precision)	$+0.02$	-0.08	-0.05	-0.03	-0.05	-0.04

- Massive ν gives oscillating residuals (peak shift); nonlinear lensing makes the tail worse $\Rightarrow$ emulator = massless ν + linear lensing.
- CosmoPower paper metric ($|\Delta C|/\sigma_{\rm CMB}$, $f_{\rm sky} = 0.4$, SO goal noise): TT 0.15–0.19σ above $\ell \approx 1600$; paper: 99% $< 0.1\sigma$ in a 5σ-Planck box, $\sim 0.2\sigma$ over the full training box.

Backup: marginalising the Fisher over the ΛCDM parameters

With $x = \delta\omega_m$, $y = \delta\theta$ ($\theta = \omega_b, h, \tau, n_s, A_s$) and $F = \begin{pmatrix} a & b^T \\ b & A \end{pmatrix}$:

$$P(x) \propto \int d^5y \, \exp\Big[-\tfrac{1}{2}\big(ax^2 + 2x\, b^T y + y^T A y\big)\Big] \propto \exp\Big[-\tfrac{1}{2}x^2\big(a - b^T A^{-1} b\big)\Big],$$

so $\sigma^2(\omega_m) = (a - b^T A^{-1} b)^{-1} = (F^{-1})_{\omega_m\omega_m}$ (Schur complement).

For N bins the same algebra with $a \to \mathrm{diag}(F_{(i)}^{\omega_m\omega_m})$ and $b \to B_{ai} = \sum_{\ell\in i} F_\ell^{a\,\omega_m}$ gives the bin covariance

$$C_{\omega_m} = \big[\mathrm{diag}(F_{(i)}^{\omega_m\omega_m}) - B^T A^{-1} B\big]^{-1}, \qquad \sigma^2(\Delta_k) = C_{kk} + C_{k+1,k+1} - 2C_{k,k+1}.$$

All per-bin sums come from cumulative sums of the per-ℓ Fisher blocks, so a binning costs microseconds.

Edge rounding (best edges on a grid): $N = 2$: 970 / 975 / 950 for steps 10 / 25 / 50, +0.0% / +0.0% / +0.3% in $\sigma(\Delta)$. $N = 3$: (870, 1230) / (875, 1225) / (850, 1250): +0.1% / +0.1% / +1.0%. Equal-information edges (943; 831, 1188) cost +1% / +7%.